

## How to Configure Whodata on Windows for Wazuh FIM

### 1- Definition

**File integrity monitoring, or FIM**, tells us what changes have occurred in our system's files and directories.

**FIM with Who Data** tells us which user and process performed the change. Like which user account made a change and which application or process made the change. Meaning the FIM with Who Data feature gives us the context of what, who, and how. Then, in cybersecurity, FIM with Who Data is very important for security analysis and incident response.

### 2- Some action FIM with who data can perform

FIM with Who Data helps to fulfill Some requirements like:

- If a critical file is changed, we can immediately see which user performed the action (for forensic analysis).
- If any malware modifies a system file, FIM with Who Data alerts us with that process name (malware detection).
- FIM with Who Data is also used for compliance requirements like PCI DSS and GDPR. Because these compliances require us to be able to track sensitive data access,

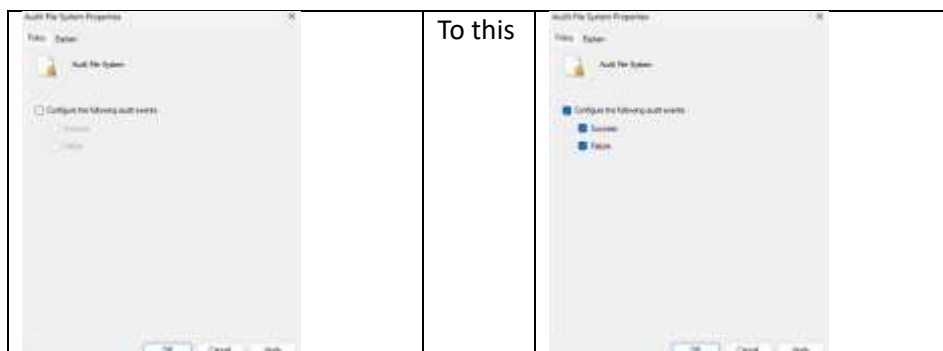
### 3- Action to perform on Windows agent machine

Before **configuring FIM with Who Data**, we have to:

#### A- enable Windows file auditing.

So go to Start and type local security policy and hit Enter. The wizard will open for us. After that:

Choose Advanced Audit Policy Configuration and click on it → click on System Audit Policy Configuration → Click on Object Access → find Audit File System → Enable it (Enable both Success and Failure).



#### B- Create folders or directories we want to track.

In our example we will have SimpleData and FinanceDocuments in C:\Users\timtc\ Downloads folder which we want to track.

#### 4- Configure FIM with Who Data on these folders so that we can collect audit logs.

For each folder, right-click, Properties → Security → Advanced → Auditing → Click on continue → click on add → click on select a principal → enter the object name to select (we choose everyone) → check Names → click ok → and assign Basic permission (we choose to assign full control, Modify, Read & execute, List folder contents, Read, Write) → click ok → apply → ok → ok.

#### 5- Apply the changes in group policy and user policy

Do: Cmd, run as administrator → type gpupdate /force

#### 6- Advanced Configuration for Comprehensive Action Coverage

To implement File Integrity Monitoring (FIM) with Who Data across all the specific actions it performs (user tracking, permission changes, sensitive files, session correlation, and compliance auditing), you must configure both the Windows agent (ossec.conf) and the Wazuh Manager (local\_rules.xml).

##### Step 1: Windows Agent Configuration (ossec.conf)

On the Windows endpoint, enable the Who Data engine within the <syscheck> block. This tells the Wazuh agent to hook into the Windows audit subsystem to capture user SIDs, process names, and executable paths.

If you have a single agent, open your Wazuh agent configuration file (typically located at C:\Program Files (x86)\ossec-agent\ossec.conf) and add or modify the syscheck section:

If you have multiple agent, Go back to your Wazuh manager SIEM → Agents management → groups → click editing on windows group, and add or modify the syscheck section:

```
<ossec_config>
<syscheck>
  <!-- Frequency of scans in seconds -->
  <frequency>3600</frequency>

  <!-- Enable Who Data real-time monitoring on a sensitive directory -->
  <directories check_all="yes" realtime="yes" report_changes="yes" whodata="yes">C:\Users\timtc\Downloads\FinanceDocument</directories>
  <directories check_all="yes" realtime="yes" report_changes="yes">C:\Users\timtc\Downloads\SimpleData </directories>

  <!-- Ignore temporary files to prevent noise -->
  <ignore>C:\Users\timtc\OneDrive\Documents\FinanceDocument\temp</ignore>
</syscheck>
</ossec_config>
```

##### Verification:

- (For single machine) Restart the Wazuh agent service via PowerShell (Restart-Service WazuhSvc) and check C:\Program Files (x86)\ossec-agent\ossec.log to ensure it successfully initializes Who Data without initialization errors.
- (For multiple agent) restart wazuh manager

**NB:** To support multiple Windows agents with different user account names, replace the hardcoded username (timtc) with the wildcard character (\*) in your path definitions. This instructs Wazuh to dynamically evaluate and monitor the target folders across any user profile directory on each endpoint.

```
<ossec_config>
<syscheck>
  <!-- Frequency of scans in seconds -->
  <frequency>3600</frequency>

  <!-- Use wildcards (*) to dynamically match any user profile folder in Downloads -->
  <directories check_all="yes" realtime="yes" report_changes="yes" whodata="yes">C:\Users\*\Downloads\FinanceDocument</directories>
  <directories check_all="yes" realtime="yes" report_changes="yes">C:\Users\*\Downloads\SimpleData</directories>

  <!-- Ignore temporary files dynamically across all user profiles -->
  <ignore>C:\Users\*\Downloads\FinanceDocument\temp</ignore>
</syscheck>
</ossec_config>
```

**For Linux past this:**

```
<!-- FIM Configuration For Linux Agents -->
<syscheck>
  <disabled>no</disabled>
  <scan_on_start>yes</scan_on_start>
  <frequency>300</frequency>
  <!-- Scans every 5 minutes as a fallback -->
  <directories check_all="yes" realtime="yes" report_changes="yes">/home/*/Desktop/SimpleData</directories>
  <directories check_all="yes" realtime="yes" report_changes="yes"
whodata="yes">/home/*/Desktop/SensitiveData</directories>
</syscheck>
```

## Step 2: Wazuh Manager Configuration (local\_rules.xml)

On the Wazuh Manager, you need custom correlation rules to parse and categorize the incoming Who Data events (such as user identity tracking, permission adjustments, and compliance violations).

Open /var/ossec/etc/rules/local\_rules.xml on your Wazuh manager and add custom rules:

```
<group name="windows, syscheck, who_data, compliance,">

  <!-- Rule 1: Detect changes to critical files capturing User and Process -->
  <rule id="100201" level="10">
    <if_sid>550</if_sid>
    <match>C:\Users\*\Downloads\FinanceDocument </match>
    <description>Critical file modified or accessed with Who Data: User '$(user)' executed process '$(extra_data)'.</description>
    <options>alert_by_email</options>
  </rule>

  <!-- Rule 2: Track permission or ACL modifications on sensitive directories -->
  <rule id="100202" level="12">
    <if_sid>554</if_sid>
    <match>C:\Users\*\Downloads\FinanceDocument </match>
    <description>Permissions or ACL changed on sensitive file/directory by user '$(user)'. Potential privilege
modification.</description>
  </rule>
```

```

<!-- Rule 3: Compliance audit logging for PCI DSS / GDPR scope -->
<rule id="100203" level="7">
  <if_sid>550, 553, 554</if_sid>
  <match>C:\Users\*\Downloads\FinanceDocument</match>
  <description>Compliance Audit: Monitored sensitive data file altered. User: '$(user)', Action logged for PCI DSS/GDPR tracking.</description>
  <group>pci_dss_10.6.1, pci_dss_11.5, gdpr_iv_35,</group>
</rule>

</group>

```

## For Linux

```

<group name="linux, syscheck, who_data, compliance,">

  <!-- Rule 1: Detect changes to critical files capturing User and Process -->
  <rule id="100201" level="10">
    <if_sid>550</if_sid>
    <match>/home/*/Desktop/SensitiveData</match>
    <description>Critical file modified or accessed with Who Data: User '$(user)' executed process '$(extra_data)'.</description>
    <options>alert_by_email</options>
  </rule>

  <!-- Rule 2: Track permission or metadata modifications on sensitive directories -->
  <rule id="100202" level="12">
    <if_sid>554</if_sid>
    <match>/home/*/Desktop/SensitiveData</match>
    <description>Permissions or attribute changed on sensitive file/directory by user '$(user)'. Potential privilege modification.</description>
  </rule>

  <!-- Rule 3: Compliance audit logging for PCI DSS / GDPR scope -->
  <rule id="100203" level="7">
    <if_sid>550, 553, 554</if_sid>
    <match>/home/*/Desktop/SensitiveData</match>
    <description>Compliance Audit: Monitored sensitive data file altered. User: '$(user)', Action logged for PCI DSS/GDPR tracking.</description>
    <group>pci_dss_10.6.1, pci_dss_11.5, gdpr_iv_35,</group>
  </rule>

</group>

```

**Verification:** Test the rule syntax on the manager using `/var/ossec/bin/wazuh-control test` and tail `/var/ossec/logs/alerts/alerts.log` after performing a test modification on the Windows agent to ensure alerts populate with the correct user and process fields.

## Testing and Verification in Wazuh SIEM

1. Go inside these docs: SimpleData and FinanceDocuments, and perform some changes (delete file and directories).

Go back to our Wazuh SIEM dashboard to see the action we performed

2. We can track this on the low severity section, click on it, and we can see the logs coming from all the connecting machines, and we can filter them there:
  - Choose our desktop machine: agent.name
  - From the edit filter: choose our rule.level (6 to 16) and save it
  - Apply different filters like: syscheck.event, rule.level, syscheck.path (for directories), rule.description, full\_log, and syscheck.audit.user.name
3. After selecting the filters, click on the to full screen view.



**WAZUH**  
Open Source Security  
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# How to Configure Whodata on Windows for Wazuh FIM

Track What, Who and How — File Integrity Monitoring with Context



"Knowledge  
Builds a Safer  
Tomorrow"

— Tims TCHOUAMOU

## 1 What is Whodata and FIM?

### Definition

File Integrity Monitoring (FIM) is an essential cybersecurity technology that monitors, detects, and alerts changes made to critical operating system files, application binaries, configuration files, and registry entries.

FIM with Who Data tells us which user and process performed the change, giving us the context of **what, who, and how**. It is very important for security analysis and incident response.

### Key Benefits

- ✓ Detect unauthorized changes
- ✓ Identify the user and process
- ✓ Monitor malware activity
- ✓ Support compliance (PCI DSS, GDPR)
- ✓ Improve incident response

"Know who changed what, and stay ahead of threats."

## 2 What Can FIM with Who Data Do?



### Forensic Analysis

Immediately see which user performed the action when a critical file is changed.



### Malware Detection

Alert when malware modifies system files with the process name.



### Compliance

Helps meet PCI DSS, GDPR and other compliance requirements by tracking sensitive data access.

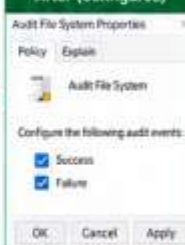
## 3 Enable Windows File Auditing (Prerequisite)

- 1 Press Windows + R, type secpol.msc and press Enter.
- 2 Navigate to: **Advanced Audit Policy Configuration** → **System Audit Policies** → **Local Group Policy Object** → **Object Access** → **Audit File System**.
- 3 Enable both **Success** and **Failure**.

### Before (Not configured)

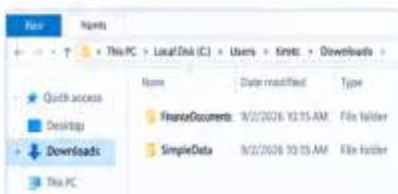


### After (Configured)



## 4 Create Folders to Monitor

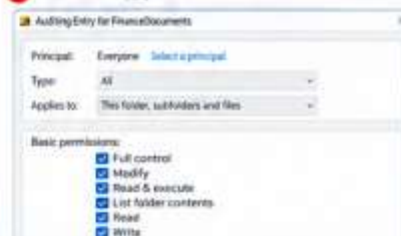
- 1 Open File Explorer and go to C:\Users\timtc\Downloads
- 2 Create the folders (example):
  - SimpleData
  - FinanceDocuments



These are the folders we will monitor for changes.

## 5 Configure Folder Auditing (Permissions)

- 1 Right-click the folder (e.g., FinanceDocuments) → Properties.
- 2 Go to the **Security** tab → **Advanced** → **Auditing**.
- 3 Click **Continue** → **Add** → **Select a principal** → enter **Everyone**.
- 4 Check **Names** → **OK**.
- 5 Select permissions (Full control, Modify, Read & execute, List folder contents, Read, Write).
- 6 Click **OK** → **Apply** → **OK**.



## 6 Apply Changes with Group Policy

- 1 Open Command Prompt as Administrator.
- 2 Run the command:

```
C:\Windows\system32> gpupdate /force
```

Administrator: Command Prompt

```
C:\Windows\system32>gpupdate /force
Updating policy...
Computer Policy update has completed successfully.
User Policy update has completed successfully.
```

✓ Group policy updated successfully!

## 7 Configure Wazuh Agent (ossec.conf) - Single Machine

- 1 Open Notepad as Administrator.
- 2 Edit: C:\Program Files (x86)\ossec-agent\ossec.conf
- 3 Add the following FIM configuration (example for one folder):

```
<!-- FIM Configuration for Windows Agents -->
<syscheck>
  <scan_on_start>yes</scan_on_start>
  <interval>60</interval>
  <disabledno>disabled</disabledno>
  <report_changes>yes</report_changes>
  <directories check_all="yes" realtime="yes">
    C:\Users\timtc\Downloads\SimpleData
  </directories>
</syscheck>
```

✓ Save the file.

## 8 Advanced Configuration (Recommended)

Use multiple directories and enable whodata to capture user and process information.

```
<ossec_config>
  <syscheck>
    <frequency>3600</frequency>
    <directories check_all="yes" realtime="yes">
      <report_changes>yes</report_changes>
      C:\Users\timtc\Downloads\FinanceDocuments
    </directories>
    <directories check_all="yes" realtime="yes">
      <report_changes>yes</report_changes>
      C:\Users\timtc\Downloads\SimpleData
    </directories>
    <ignore>C:\Users\timtc\OneDrive\Documents\
      FinanceDocument\temp</ignore>
  </syscheck>
</ossec_config>
```

Use wildcards for multiple users:  
C:\Users\\*\Downloads\FinanceDocuments  
C:\Users\\*\Downloads\SimpleData

## 9 Wazuh Manager Rules (local\_rules.xml)

- 1 On the Wazuh manager, open: /var/ossec/etc/rules/local\_rules.xml
- 2 Add custom rules to parse and categorize Whodata events.

```
<rule name="windows_syscheck_whodata_compliance">
  <rule id="390201" level="18">
    <if_sid>3104</if_sid>
    <match>C:\Users\timtc\Downloads\FinanceDocuments\match</match>
    <description>Critical file modified or accessed with
      Who Data: User {user} executed process {extra_data}.
    </description>
    <options>alert_by_email</options>
  </rule>
```

## 10 Restart Services

- Restart Wazuh agent (single machine):

```
PS C:\> Restart-Service WazuhSvc
```

- Or restart Wazuh manager (for multiple agents):

```
sudo systemctl restart wazuh-manager
```

## 11 Verify and Test

- 1 Make a change in the monitored folder (e.g., create, modify, or delete a file).
- 2 Check logs in Wazuh dashboard → **Security events**.
- 3 Confirm the event shows the user and process that made the change.

| wazuh | Time                  | Agent    | Rule ID | Description                                                  |
|-------|-----------------------|----------|---------|--------------------------------------------------------------|
|       | Sep 2, 2026, 10:25:32 | win-pc01 | 190201  | Critical file modified<br>User: timtc   Process: notepad.exe |

## 12 Final Notes

- ✓ Use wildcards for multiple Windows users.
- ✓ Monitor sensitive folders (e.g., Downloads, Documents).
- ✓ Customize rules for your environment.
- ✓ Supports malware detection, forensics, and compliance.
- ✓ Keep your configuration simple and scalable.

"Knowledge Builds a Safer Tomorrow"  
— Tims TCHOUAMOU



**Tims TCHOUAMOU**  
Cybersecurity Enthusiast | SOC Content Creator



Open Source Security  
for a Safer World

